

Project Report: Solar Installation & Performance Analysis Dashboard

1. Overview

This is a **Power BI report** built to track and analyze rooftop/utility solar installations — covering installation volumes, capacity, generation output, costs, and subsidies across states, cities, vendors, and panel types. The report contains **2 pages**, both titled "*Solar Installation & Performance Analysis Dashboard*", and is built on a single data table: **Clean_Solar**.

2. Data Model

The report draws from one core table, Clean_Solar, containing the following fields:

Category	Fields
Identifiers	Installation_Id, Vendor, Panel_Type, Category
Location	State, City
Time	Installation_Date, Year
Capacity & Installations	Capacity_Kw, Total_Capacity, Total_Installation
Generation	Monthly_Generation_Units, Total_Generation_Units
Cost & Subsidy	Avg_Cost_Per_Kw, Net_Cost, Central_Subsidy, State_Subsidy, Total_Subsidies, Subsidy_%

Total_Installation, Total_Capacity, Total_Generation_Units, Avg_Cost_Per_Kw, Net_Cost, and Subsidy_% appear to be calculated measures (aggregations/ratios) rather than raw columns.

3. Page 1 — Installation & Capacity Overview

KPI Cards: Average Cost per Kw, Capacity (Kw), Central Subsidy, State Subsidy, and an installation count.

Visuals:

- **Line chart** — Installation trend over time (Installation_Date vs. Total_Installation)
- **Column chart** — "Top 5 Cities by Installations"
- **Scatter chart** — "State Vs Capacity" (installation count, capacity, and monthly generation by state)
- **Clustered column chart** — "Total Installation & Capacity by State"

Filters (slicers): Capacity_Kw, State, Category, Vendor, City, Year

4. Page 2 — Generation, Cost & Subsidy Analysis

KPI Cards: Installation count, Monthly Generation Units, Net Cost, Subsidy %, Total Subsidies.

Visuals:

- **Pie chart** — "Total Generation Units by Panel Type"
- **Clustered bar chart** — "Total Installations by Category"
- **Column chart** — "No. Of Installations by Vendor"
- **Line chart** — "Monthly Generated Units" trend over time

Filters (slicers): City, Panel_Type, Vendor, State, Year, Category

5. Key Capabilities

- Cross-filtering via 6 slicers per page (geography, vendor, panel type, category, capacity, year)
- Financial tracking: cost per kW, net cost, and central/state subsidy breakdowns
- Performance monitoring: monthly and total generation output segmented by panel type
- Comparative analysis: installations and capacity benchmarked across states, cities, and vendors

6. Notes / Limitations

The report's underlying data model is stored in a compressed binary format inside the .pbix file, so I could read the **structure** (pages, visuals, fields, chart types, titles) but not the **actual row-level data or current metric values** — that requires opening it in Power BI Desktop or Power BI Service. If you'd like, I can also look at row counts/sample data if you export the underlying dataset (e.g., as CSV or Excel).